

Pankaj Khatti

Second Year Postgraduate
Department of Mechanical Engineering

khattipankaj@iitgn.ac.in
+91 9460177451

Education

Degree	Institution	CPI/%	Year
M.Tech (Mechanical)	IIT Gandhinagar	8.29	2022-Present
B.Tech (Mechanical)	MDU, Rohtak	80.13%	2016-2020
Class XII	National Public School, HMH	84.0%	2015-2016
Class X	National Public School, HMH	8.8	2013-2014

Experience

- **Operating Engineer Trainee, Quality Assurance, Caparo Engineering India Limited;** [July 2021-July 2022]
 - Supplier Quality Engineer in the Quality Assurance department.
 - Responsible for incoming and supplier quality for **254** purchased parts from **11 suppliers**, evaluating quality, delivery, and defect response. Enhance performance, and conduct annual plant audits for supplier ranking.
 - Responsible for monthly **rework and rejection data**, addressing issues, and managing customer complaints, defects, and process enhancements. I conduct **raw material testing** and lead **SQE** during **IASTF audits**.
 - Proficient in using the industry-standard **7 QC** tools for data analysis and problem-solving. Skillful in employing methodologies like **Why-Why analysis, 4M, and 5S** to enhance processes and identify root causes. Experienced in applying core quality management tools like **APQP, FMEA, SPC, MSA, and PPAP** to ensure product and process excellence and provide parts as per **GD&T** and customer requirements.
 - Implement **9 Kaizen** ideas for process improvement and achieve **3 months of zero customer complaints**.

Internship

- **Graduate Apprentice trainee, Hindustan Petroleum Cooperation Limited;** [April 2021-July 2021]
 - As a GAT at HPCL Rajasthan Refinery Limited, I conducted site inspections, gathered construction data, and reported findings. I also liaised with local authorities for regulatory compliance and permissions
 - Responsible for preparing the Engineering store tender report for the marketing terminal. Gained knowledge about various fuels and chemicals from crude oil refining.
- **Techno commercial Engineer, Starmoon Consulting Services Pvt. Ltd** [Jan 2020-April 2020]
 - Provide SOLID EDGE training, and consulting for CAD software to industry professionals.
 - Give training on the use of Microsoft Office to small-scale industries for work online.
 - Learn teaching skills and understand the problems of small-scale industries and help them to compete with the fast-growing big industries.

Projects

- **Flexible Manipulator, Smart Robotics and Manufacturing Lab, IIT Gandhinagar** [Jan 2023-April 2023]
 - Designed and prototyped a versatile vertical flexible manipulator for experiments, with adaptable, portable features. Executed efficient manufacturing processes, ensuring project success
- **Control Lyapunov controller on Inverted Pendulum, IIT Gandhinagar** [Jan 2023 - April 2023]
 - Design the nonlinear controller by using the control Lyapunov function by considering its dynamics and doing the simulation on Matlab.
- **The Khaputli Show, ES 408 Mechatronics, IIT Gandhinagar** [Aug 2022-Nov 2022]
 - Led the successful development of a captivating dancing puppet showcasing a 3 Degree of Freedom (DOF) mechanism. Integrated linear actuators and pulley arrangements with Arduino-based control systems, seamlessly merging mechanical and electrical components.

Positions of Responsibility

- **Teaching Assistant, ME 106 and ES 115 Undergrad Course, IIT Gandhinagar** [Aug2022-April 23]
 - Instructed undergraduate students on sheet metal operations and CNC lathe and milling machine usage, delivering comprehensive theoretical and practical lessons.
 - Guided **6 teams of 10 students each** for a course project, providing mentorship, facilitating project planning, encouraging teamwork, and ensuring adherence to timelines and deliverables.

Achievements

- Ranked **1st** in 3rd semester, **4th** in 4th semester, and **8th** in 2nd semester, MDU merit list.
- Served as a team leader in the **2nd round of SMART INDIA HACKATHON (SIH)**,
- Represented the school volleyball team in the **CBSE cluster**.

Skill Summary

- **Languages:** Python, Matlab
- **Tools:** Solid Works, SAP, Solid Edge, Fusion 360

Extra-Curricular Activities

- Event organizer in the college's annual feast.
- Actively participate in Scout and guide in school.
- Annually, I volunteer for water-feeding initiatives in my village, ensuring access to clean water for residents.